SCIENCE DIALOGUE · サイエンス・ダイアログ

Where do your eyes go?

Teaching a computer to look at the world the way you do

コンピューターに「人の見方」を教える

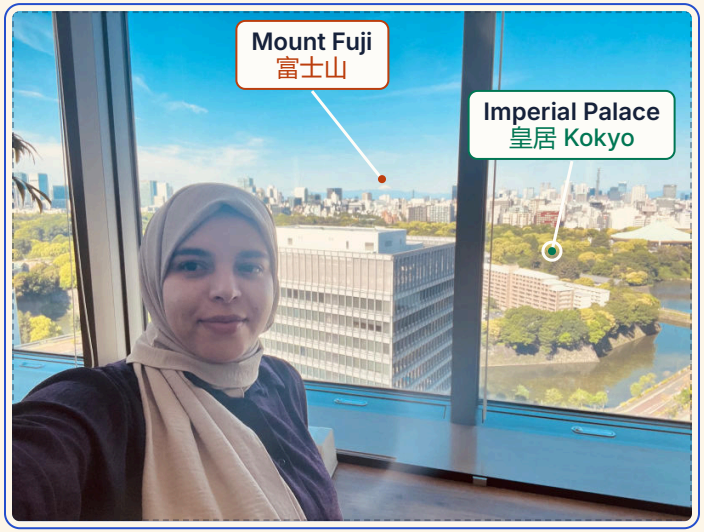
Dr. Zaineb IBORK · JSPS Postdoctoral Fellow, National Institute of Informatics (NII)

· Utsunomiya Girls' High School · 26 October 2026



Can you guess where I come from?

私がどこから来たか分かりますか？



- I am from Morocco, in North Africa.
- I work in Japan, at the National Institute of Informatics.
- I study 3D shapes: how computers see them, and how YOU see them.
- Today: no hard words. Stop me any time.

WORDS guess 当てる country 国 shape 形

01 · LANGUAGES · 言語

How do you say “hello” in Morocco?

モロッコで「こんにちは」は何と言いますか？

	DARIJA (ARABIC)	AMAZIGH	FRENCH	日本語
Hello	Salam alaykum · السلام عليكم	Azul · ⴰⵣⵓⵍ	Bonjour	こんにちは
Thank you	Shukran · شكرا	Tanmirt · ⵜⴰⵏⵎⵔⵜ	Merci	ありがとう
My name is…	Smiti… · سميتي	Isem inu… · ⵉⵙⵎ ⵉⵏⵓ	Je m'appelle…	私の名前は…

Two official languages, Arabic and Amazigh. French and English at school.
Many of us change language several times a day.

WORDS language 言語 alphabet 文字 official 公用の

Source: Constitution of Morocco (2011), art. 5; Ethnologue

01

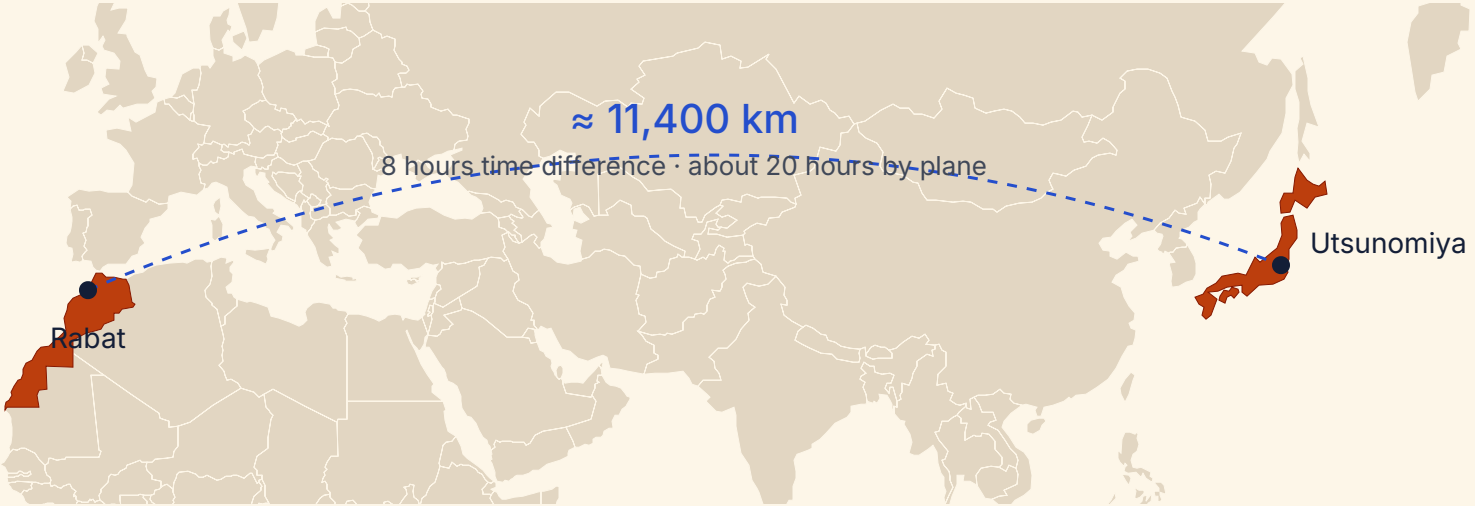
Morocco and Japan

How different are we, really?

モロッコと日本：私たちはどれくらい違う？

Where is Morocco on the map?

モロッコは地図のどこにありますか？



WORDS map 地図 distance 距離 Africa アフリカ time difference 時差

Source: Distances calculated on Google Maps (great-circle)

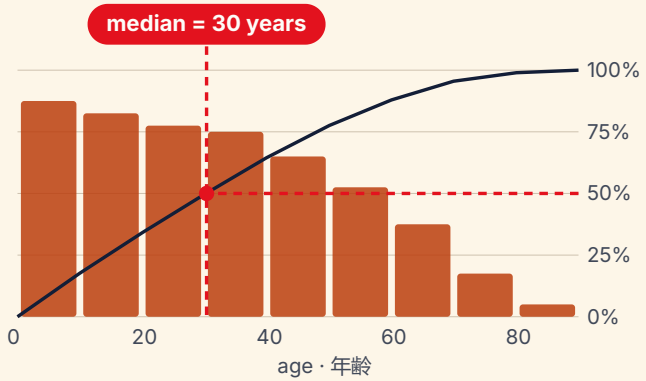
How many people live in each country?

それぞれの国に何人住んでいますか？

Bars: % of people in each age group (ヒストグラム). **White line**: we add them up. Where it reaches **50%**, read the age: that is the median (中央値).

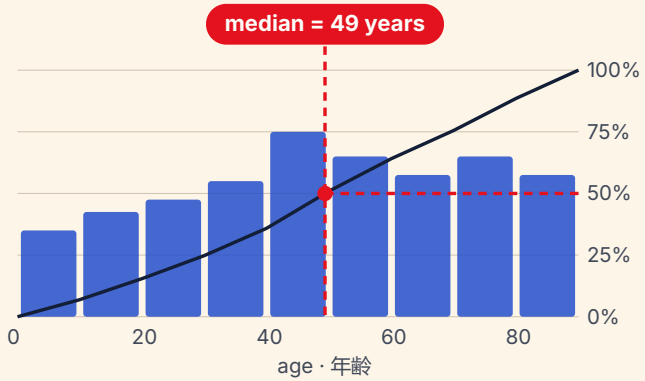
MOROCCO · モロッコ

37 million



JAPAN · 日本

124 million



WORDS population 人口 million 百万 average age 平均年齢 median 中央値

Source: UN, 2024

What is the capital of Morocco?

モロッコの首都はどこですか？



MOROCCO · RABAT · ラバト

Not Casablanca: Rabat!

An old kasbah next to the Atlantic Ocean, and a new tower behind it.



JAPAN · TOKYO · 東京

About 37 million people in the Tokyo area

The same number of people as all of Morocco.

WORDS capital 首都 ocean 海洋 old town 旧市街

Source: Photos: Wikimedia Commons

Which values do Morocco and Japan share?

モロッコと日本はどんな価値観を共有していますか？

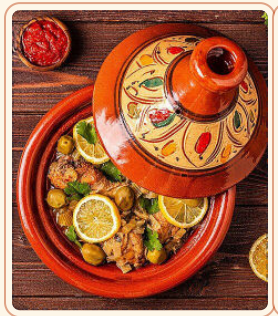
ISLAM · イスラム教		JAPAN · 日本
Seek knowledge all your life: be better every day 一生学び続ける	⟷	**Kaizen: small improvements, every day** 改善 カイゼン
“Saving one life is like saving all humanity” (Qur’an 5:32) 命と人権の尊重	⟷	**Respect for life and for every person** 命を大切に
Work is a trust (amana) and a way to worship 仕事は信託	⟷	**Commitment to work: the shokunin spirit** 職人の心
Itqan: do your work well. Ihsan: do it with excellence, as if God sees you イトカーン < イフサーン	⟷	**Doing it right, even when nobody is watching** 丁寧・誠実

WORDS value 価値観 purpose 目的 excellence 卓越 respect 尊重

Source: Qur’an 5:32; Hadith of Jibril (Sahih Muslim); M. Imai, Kaizen, 1986

What would you eat for lunch in Morocco?

モロッコの昼食に何を食べますか？





TAJINE ↔ 鍋 NABE

Cooked slowly in a clay pot

Just like a donabe in Japan.



COUSCOUS · クスクス

The Friday family meal

Tiny grains of wheat, eaten from one big plate together.





MINT TEA ↔ 抹茶 MATCHA

We both love green tea

Ours has fresh mint and a sugar.

WORDS dish 料理 clay pot 土鍋 mint ミント sugar 砂糖

Source: Photos: Wikimedia Commons and personal photos

What would you wear to a Moroccan wedding?

モロッコの結婚式で何を着ますか？



DJELLABA · ジェラバ

Every day, with a hood

A long, comfortable robe for men and women.



KAFTAN ↔ 着物 KIMONO

For weddings and celebrations

Silk, gold thread, and hours of handwork.



BABOUCHE ↔ 下駄 GETA

Special shoes for special days

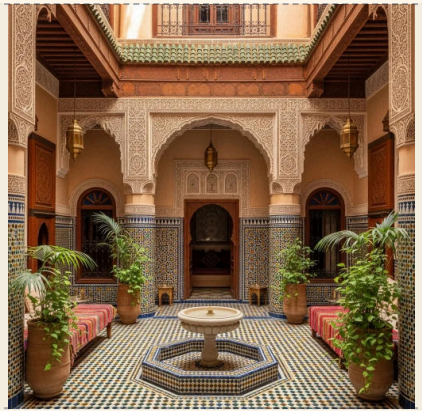
Embroidered babouches with a kaftan, geta with a yukata.

WORDS clothes 服 wedding 結婚式 celebration お祝い

Source: Photos: Wikimedia Commons and personal photos

What is inside a traditional house?

伝統的な家の中には何がありますか？



MOROCCO · RIAD · リヤド

A hidden courtyard with a fountain

Plain walls outside. Zellige tiles, carved plaster and sky inside.



JAPAN · WASHITSU · 和室

Tatami, wood and sliding doors

Simple and calm. In both houses: shoes off, and we sit low.

WORDS courtyard 中庭 tile タイル sliding door ふすま・障子

Source: Photos: Wikimedia Commons

TODAY'S WORDS · 今日の言葉

Which fifteen words will help you today?

今日役に立つ15の言葉は？

1 **3D model / mesh**
3Dモデル／メッシュ
a shape built from many small triangles

2 **triangle / vertex**
三角形 / 頂点
the small pieces a 3D shape is made of

3 **saliency**
顕著性
the part that catches your eyes first

4 **attention**
注意
where the eyes and the brain decide to go

5 **eye-tracking**
視線計測
a camera that records where you look

6 **feature**
特徴量
a clue the computer can measure with a number

7 **curvature**
曲率
how much a surface bends

8 **algorithm**
アルゴリズム
a “recipe” a computer can follow

9 **simplification / compression**
簡略化 / 圧縮
making a heavy file lighter

10 **quality assessment**
品質評価
does a 3D model look good to a human?

11 **subjective test**
主観評価
asking real people, and counting

12 **experiment**
実験
testing an idea instead of believing it

13 **generalization**
汎化
it also works on new, unknown objects

14 **cultural heritage**
文化財
temples and statues, scanned in 3D

15 **curiosity**
好奇心
asking Why? How? When?

02

My path to science

Why did I become a scientist?

なぜ科学者になったの？



Why did I ask “why?” so often?

なぜ私はそんなに「なぜ？」と聞いたのでしょう？

- Why does water take the shape of the glass?
  なぜ水はコップの形になるの？
- Why can I see my friend from very far away, but a computer cannot?
  なぜ遠くの友達が見えるのに、コンピューターには見えないの？
- Why does one cartoon look real, and another look fake?
  なぜあるアニメは本物っぽく、別のアニメは偽物っぽく見えるの？

WORDS scientist 科学者 question 質問

How many times did I change my clothes?

私は何回服を着替えたでしょう？

Fascinated by the natural world
自然が大好き



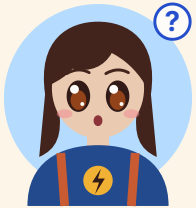
Until middle school

School · 学校

Good at all subjects and sports

→ **Curious child**
好奇心旺盛な子

How do things work? Electricity
仕組み・電気



2011 – 2012

Bac, electrical technology

Technical high school, Kenitra

→ **Technology student**
技術系の高校生



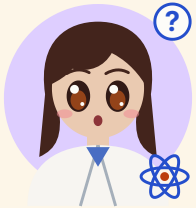
2012 – 2014

DUT, EST Fès

Electronics and industrial computing

→ **Senior technician**
技術者

Back to the physical world
物理の世界へ



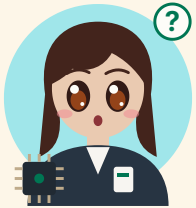
2014 – 2018

Bachelor in physics

Ibn Tofail University, Kenitra

→ **Physicist**
物理学者

Back to the technical world
技術の世界へ



2018 – 2020

Master, embedded electronics

Ibn Tofail University, Kenitra

→ **Engineer**
エンジニア

Curious about advanced technologies
最先端技術

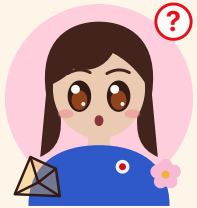


2021 – 2025

PhD, computer science and AI

Caen, France and Kenitra

→ **Doctor**
博士



2026 – 2027

Postdoc, NII Tokyo

JSPS fellowship

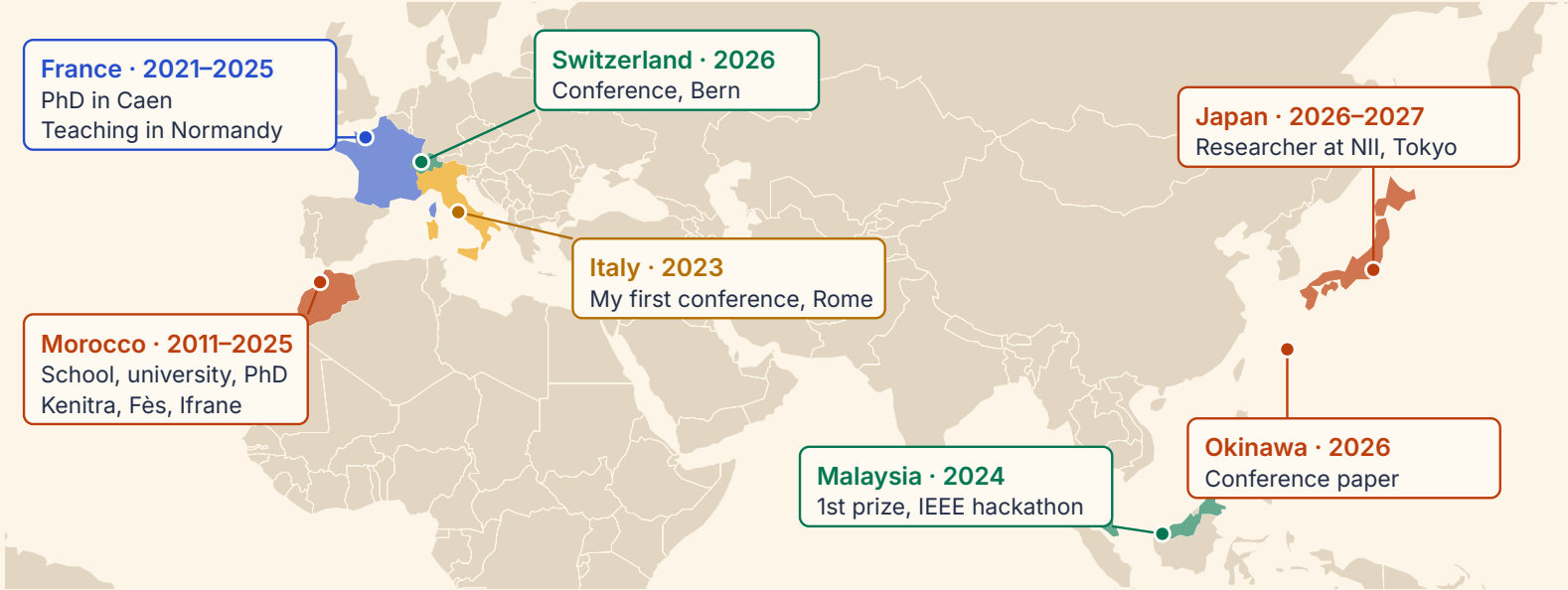
→ **Researcher in Japan**
研究者

WORDS technician 技術者 bachelor 学士 master 修士 PhD 博士

Source: Z. Ibork, curriculum vitae, 2026

Where has my curiosity taken me?

好奇心は私をどこへ連れて行ったでしょう？



WORDS country 国 conference 学会 prize 賞

Source: Z. Ibork, curriculum vitae, 2026

03

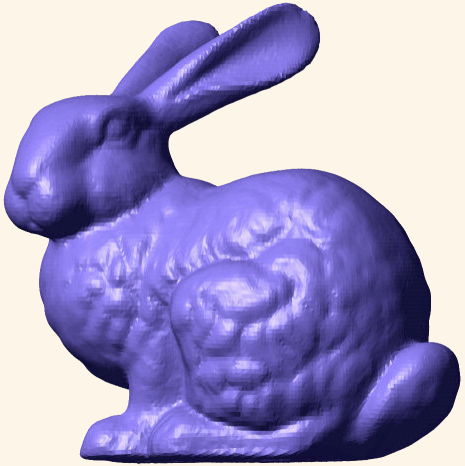
My research

Can a computer see the way you do?

コンピューターは人と同じように見られる？

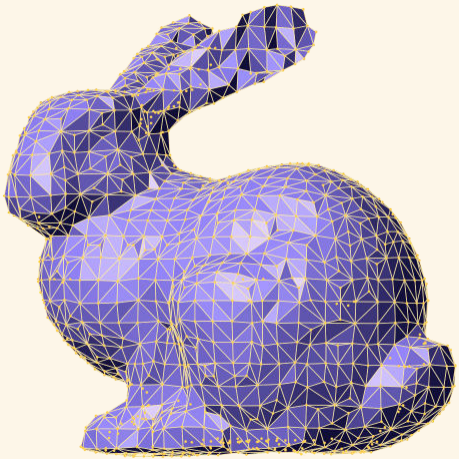
What is a 3D model made of?

3Dモデルは何でできていますか？



What you see

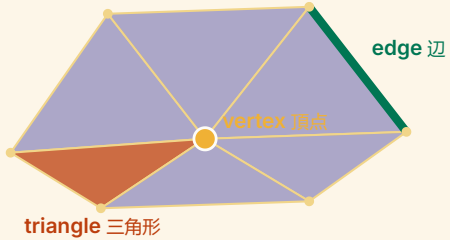
見た目



What the computer stores

メッシュ

ZOOM · 拡大



One vertex = 6 numbers · 6つの数字

x	y	z	R	G	B
0.42	1.08	−0.27	145	132	217

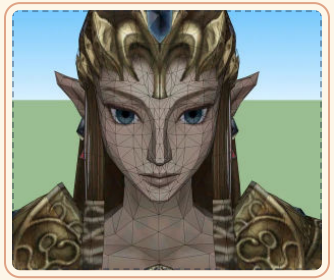
Position · 位置 | **Color · 色**

WORDS vertex 頂点 edge 辺 triangle 三角形

Source: Stanford 3D Scanning Repository, 1994

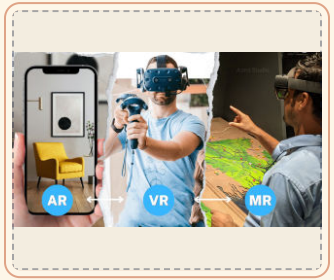
Where have you already seen a 3D mesh?

3Dメッシュをどこで見たことがありますか？



GAMES · ゲーム

Every character is a mesh.



AR · VR · MR

Virtual objects in your room.



3D SCAN · 3Dスキャン

Your face, measured in 3D.



3D PRINTING · 医療

Bones, cars, buildings, jewels.



HERITAGE · 文化財

Temples saved forever.

WORDS application 応用 scan スキャン heritage 文化財

Source: Z. Ibork, PhD defence, Université de Caen Normandie, 2025

03

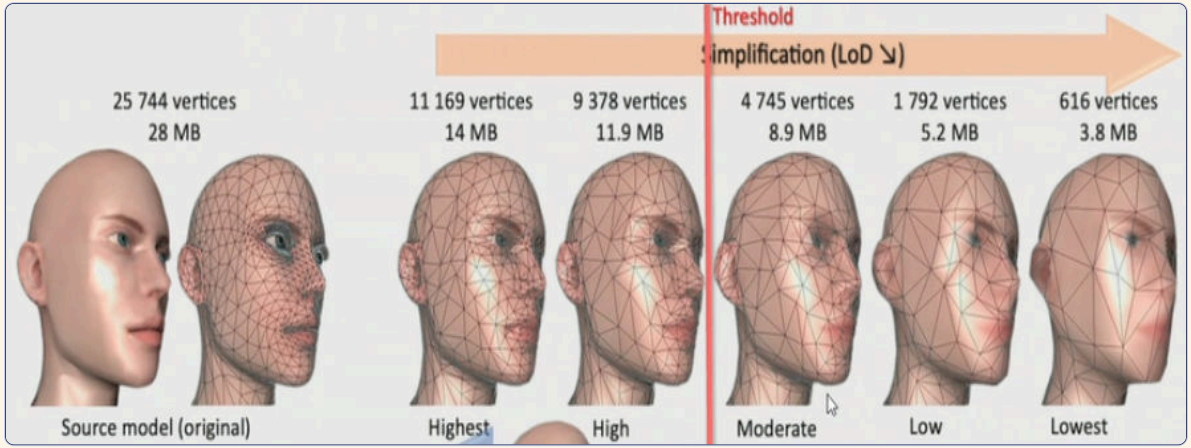
Can your eyes judge the quality of a 3D mesh?

あなたの目で3Dメッシュの品質を評価できますか？

Let’s find out: you will be the judges・みなさんが審査員です

Where would you draw the red line?

どこに赤い線を引きますか？



Original · 元のモデル: 25,744 vertices, 28 MB.
頂点25,744個、28MB

Before the line: half the size, the same look.
線の前：半分のサイズ、同じ見た目

After the line · しきい値: the face starts to break.
線の後：顔が崩れ始める

WORDS vertex 頂点 threshold しきい値 level of detail 詳細度

Source: Z. Ibork, PhD defence, Université de Caen Normandie, 2025

ACTIVITY · 形だけで当てよう

Which armadillo is the original?

どのアルマジロがオリジナルですか？



A



B



C

WORDS original オリジナル shape 形

Source: LIRIS/EPFL General-Purpose dataset, Lavoué et al., SPIE 2006

ANSWER · 答え

Did you find the original?

オリジナルを見つけられましたか？



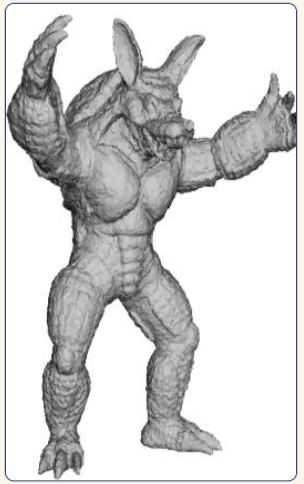
A · Original

元のモデル



B · Smoothed

平滑化



C · Noise

ノイズ

WORDS　smoothing 平滑化　noise ノイズ

Source: LIRIS/EPFL General-Purpose dataset, Lavoué et al., SPIE 2006

ACTIVITY · 品質を当てよう

What score would you give the copy?

コピーに何点をつけますか？



Original

元のモデル



Copy

コピー

The original is perfect: **100**.

Your score for the copy, from 0 to 100?

コピーの点数は？（0〜100）

Show it with your fingers: 1 finger = 10.

WORDS score 点数 original オリジナル copy コピー

Source: Samurai mesh, CMDM dataset, Nehmé et al., IEEE TVCG 2021

ANSWER · 答え

What did real people say?

実際の人々は何と答えたでしょう？



90.12

MOS · 平均評価



15.53

MOS · 平均評価

MOS = Mean Opinion Score: the average of many people's scores.

Even the original gets 90, not 100.

This dataset needed 11,520 human scores.

WORDS average 平均 opinion 意見 subjective test 主観評価

Source: Nehmé et al., IEEE TVCG 27(3), 2021; Z. Ibork, PhD defence, Université de Caen Normandie, 2025

One million 3D models: can people score them all?

100万個の3Dモデル：人が全部採点できますか？

No. Too slow, too expensive. **So we build algorithms · だからアルゴリズムを作る**

PEOPLE · 人間

主観評価

Ask people



✓ The truest answer.

✗ Slow and expensive.

ALGORITHMS · アルゴリズム

フルリファレンス

Compare with the original



+



✓ Fast and simple.

✗ Often, there is no original.

ノーリファレンス

No original needed

no original

+



✓ Works everywhere, like your eyes.

→ My PhD models · 私の研究

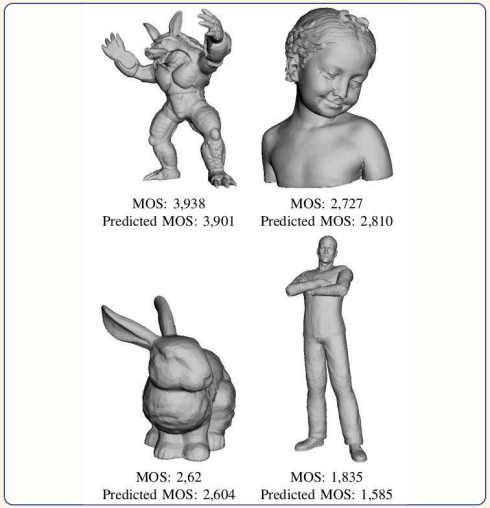
WORDS algorithm アルゴリズム original 元のモデル no reference 参照なし

Source: Ibork et al., IEEE Access 12, 2024; Z. Ibork, PhD defence, Université de Caen Normandie, 2025

MY RESULTS · 結果

Can my model score like humans?

私のモデルは人間のように採点できますか？



MESH	HUMANS · 人間	MY MODEL · AI
Armadillo	3.94	3.90
Bimba	2.73	2.81
Bunny	2.62	2.60
Man	1.84	1.59

Almost the same, without the original.

WORDS predict 予測する result 結果 close 近い

Source: Ibork et al., IEEE Access 12, 2024 (LIRIS, IEETA, UWB datasets)

Why do game characters look more real every year?

なぜゲームのキャラクターは毎年リアルになるのでしょう？



Year · 年	1996	2000	2003	2006	2008	2013	2018
Polygons · ポリゴン	< 500			10,000	32,000		> 100,000
Game size · 容量	< 2 GB			7 GB		15 GB	33 GB

Screen · 画面



WORDS polygon ポリゴン realistic リアルな size 容量

Source: infograds.com; Z. Ibork, PhD defence, 2025

03

Where do your eyes go first on a 3D mesh?

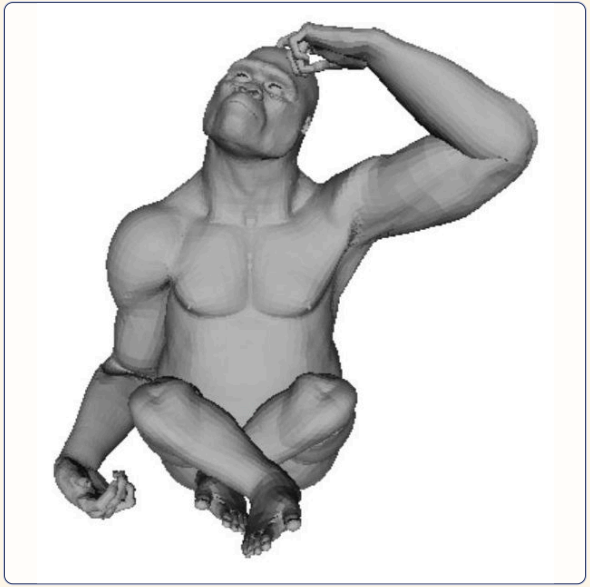
3Dメッシュを見るとき、あなたの目は最初にどこへ行きますか？

Let’s find out: visual saliency · 視覚的顕著性

SALIENCY TEST · 顕著性テスト

Where did your eyes go first on this 3D model?

この3Dモデルで、あなたの目は最初にどこを見ましたか？



Look for two seconds.

2秒間見てください。

Face? Hands? Feet? Chest?

顔? 手? 足? 胸?

Point at the first place you looked.

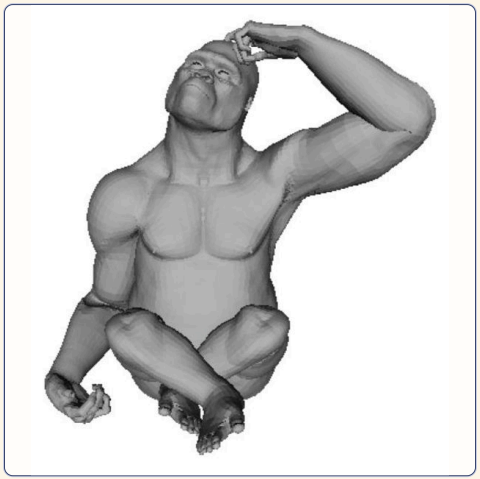
最初に見た場所を指さしてください。

WORDS look 見る first 最初に attract 引きつける point 指さす

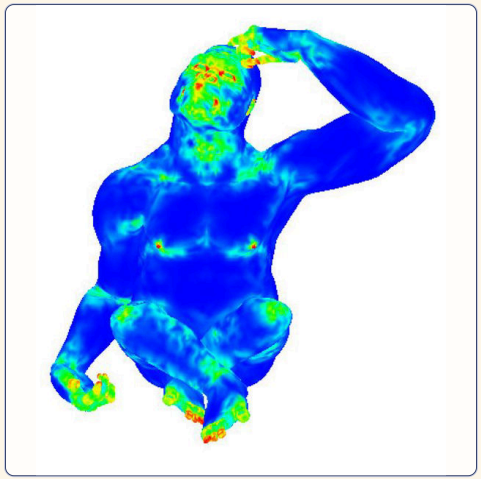
Source: Z. Ibork, PhD defence, Université de Caen Normandie, 2025

Did you look at a red place?

赤い場所を見ましたか？



3D model · 3Dモデル



Saliency map · 顕著性マップ

Saliency · 顕著性: what catches your eyes first · 最初に目を引きつける部分



Red: face, fingers, toes. Eyes go here.
赤：顔、指、つま先。目が行く場所。

Blue: smooth chest. Almost nobody looks.
青：なめらかな胸。ほとんど見ない場所。

WORDS saliency map 顕著性マップ　attention 注意　red 赤　blue 青

Source: Z. Ibork, PhD defence, Université de Caen Normandie, 2025

04

How can a computer learn to see like you?

コンピューターはどうやって人のように見ることを学ぶのでしょう？

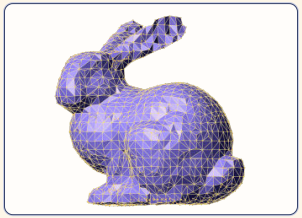
From your eyes to artificial intelligence · 人の目から人工知能へ

How will the computer judge the mesh?

コンピューターはどうやってメッシュを評価するのでしょう？

1 · 3D mesh

3Dメッシュ



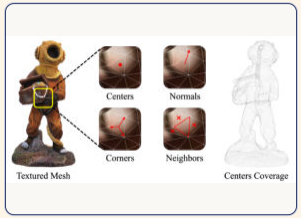
Vertices, edges, triangles.

頂点・辺・三角形

→

2 · Clues

特徴量



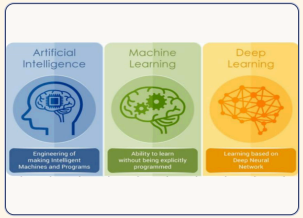
Position, color, curvature.

位置・色・曲率

→

3 · AI

人工知能



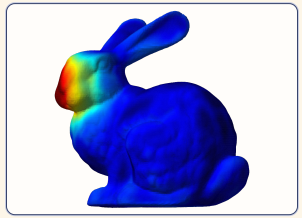
A neural network reads the clues.

ニューラルネットワーク

→

4 · Answer

答え



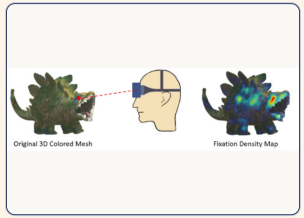
A quality score and a saliency map.

品質スコアと顕著性マップ

→

5 · Check with humans

人間と比べる



Close to people? It learned.

人に近ければ、学習成功

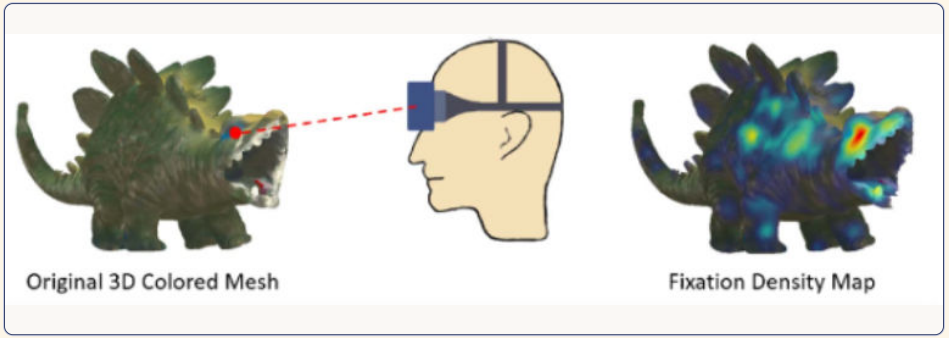
No original needed: the computer judges the mesh alone, like your eyes · 元のモデルは不要

WORDS feature 特徴量 curvature 曲率 predict 予測する

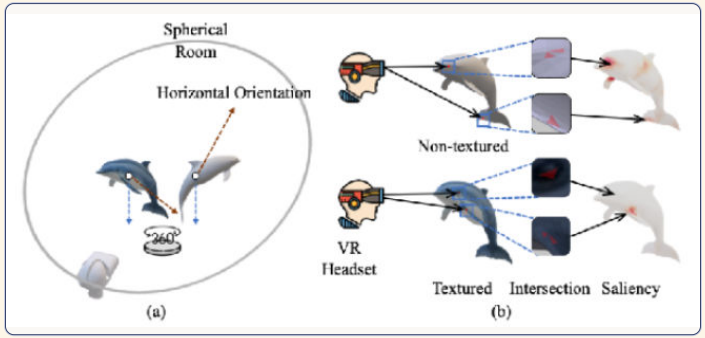
Source: Ibork et al., IEEE Access 2024; Ibork & Sugimoto, 2026

How does a headset see where you look?

ヘッドセットはどうやってあなたの視線を知るのでしょう？



Ding et al., IEEE Transactions on Multimedia, 2024



Zhang et al., CVPR 2025

1 **Wear a VR headset**
ヘッドセット
A tiny camera watches your eyes.

2 **Your eyes stop**
注視点
Each stop is a point on the 3D model.

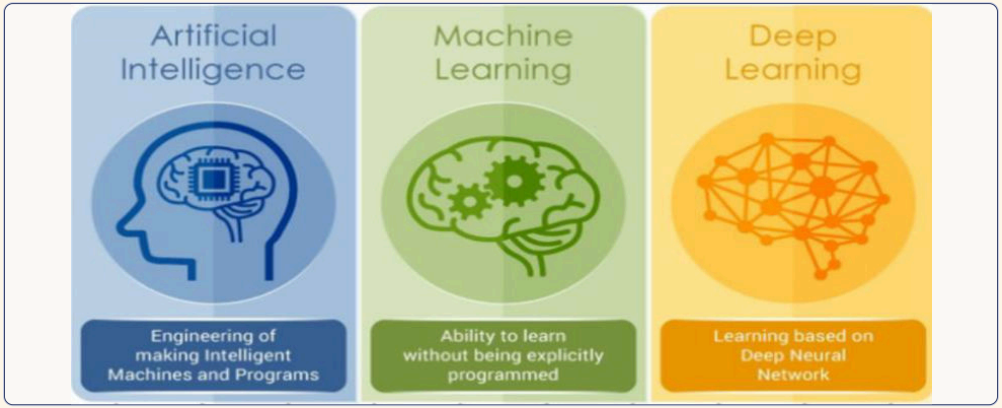
3 **Add many people**
注視密度マップ
Red: many looked. Blue: few looked.

WORDS headset ヘッドセット fixation 注視 density map 密度マップ

Source: Ding, Chen, Lin, Chen, IEEE TMM 2024; Zhang, Zhu, Min, Zhai, CVPR 2025

What is the motor of my research?

私の研究の原動力は何でしょう？



AI
人工知能

Machine learning
機械学習

Deep learning
深層学習

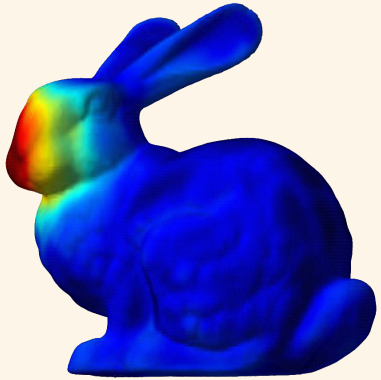


Humans **versus** machines? No: machines learn **from** humans.

Source: Towards Data Science; Z. Ibork, PhD defence, Université de Caen Normandie, 2025

What if we delete only where nobody looks?

誰も見ない場所だけを消したらどうなる？

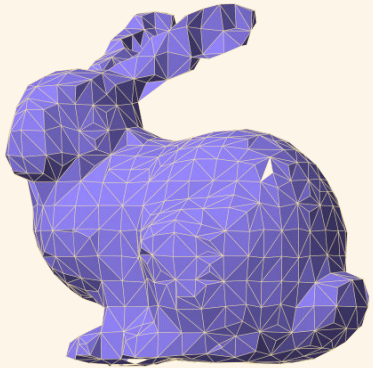


1 · Saliency map

顕著性マップ

Red: everyone looks here. Blue: nobody looks.

赤：みんなが見る場所。青：誰も見ない場所。

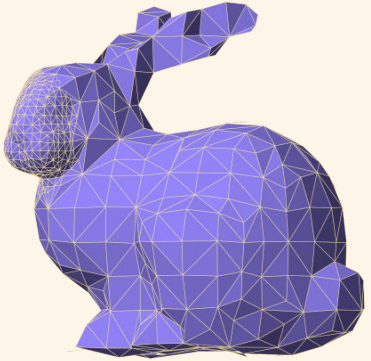


2 · Same everywhere

均一な簡略化

1,440 triangles, spread evenly. The face breaks.

1,440個を均等に。顔が崩れる。



3 · Guided by saliency

顕著性による簡略化

Same budget: many triangles on the face, few on the back.

同じ数で、顔に多く、背中に少なく。

Saliency keeps the quality, in an optimised way · 顕著性で品質を守り、最適化する

WORDS saliency map 顕著性マップ　keep 残す　optimise 最適化する

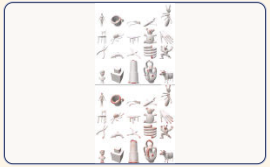
Source: Stanford Bunny; illustration of Ibork & Sugimoto, ACIVS 2026

How does one idea become science?

一つのアイデアはどうやって科学になるのでしょう？

1 — 2021

Dataset
データセット

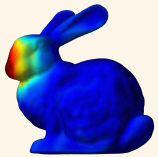


3D models with human scores and eye data.

人の評価と視線データ付きの3Dモデル

2 — 2021

Hypothesis
仮説

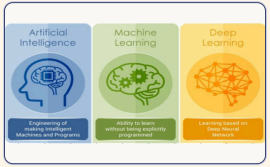


“Delete only where nobody looks.”

「誰も見ない場所だけを消す」

3 — 2022–23

Experiment
実験



Train the neural network: the AI motor.

ニューラルネットワークを学習させる

4 — 2023–24

Verification and test
検証・テスト

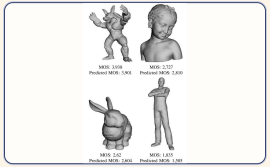


New 3D models it never saw. Compare with humans.

見たことのない3Dモデルで人と比べる

5 — 2024

Results
結果



Almost the same answers as humans.

人間とほぼ同じ答え

6 — 2024–26

Share
発表



Papers and talks, in English.

英語で論文と発表

The same six steps work for your science club and your homework. · 部活や宿題でも同じ6ステップが使えます。

What did you just do?

今、何をしたのでしょう？

You were my experiment

みなさんが私の実験でした

Two versions, real humans, count the votes. Your hands are my data.

2つのバージョン、本物の人間、票を数える。みなさんの手が私のデータです。

Why humans, not maths?

なぜ数式ではなく人間なの？

The question is “does it look good to a human?” Only you can answer.

問題は「人間の目にきれいに見えるか？」。答えられるのはみなさんだけです。

Eye-tracking

視線計測

A camera follows your eyes 60 times per second. I compare it with my red map.

カメラが1秒に60回、目の動きを追います。それを私の赤いマップと比べます。

WORDS subjective test 主観評価 eye-tracking 視線計測 data データ

Source: Ibork et al., IEEE SMC 2024; IEEE Access 2024

STEP 6 OF 6 · 発表

Why does English matter so much?

なぜ英語はそんなに大切なのでしょう？

1 · Write

A paper in English. Months of work.

2 · Get judged

Strangers try to break it. Often they say no.

3 · Present

200 people from 30 countries, in English.

4 · Be built on

Someone stands on my shoulders.

English is not the goal. English is the door.

WORDS　paper 論文　conference 学会　colleague 同僚

IF YOU REMEMBER ONLY ONE SLIDE · まとめ

What three things will you take home today?

今日、持ち帰る3つのことは何ですか？

1

Three words · 3つの言葉

3D mesh · 3Dメッシュ: a shape made of triangles.
三角形でできた形

Quality · 品質: does it still look good to you?
あなたの目にきれいに見える？

Saliency · 顕著性: what catches your eyes first.
最初に目を引きつける部分

2

A real experiment · 実験

You pointed, you scored, you voted. Your answers were real data.

指さして、点数をつけて、投票した。みなさんの答えは本物のデータです。

3

Curiosity + English · 好奇心 + 英語

Together, they took a girl from Morocco to Japan. They can take you anywhere.

この二つが、モロッコの女の子を日本へ連れてきました。みなさんもどこへでも行けます。

And two secrets · 二つの秘密:

Mistakes · 間違い: my code crashes every day. A wrong result is data.
間違った結果もデータです。

Stubbornness · 粘り強さ: keep asking when everyone else got bored.
みんなが飽きても、問い続けること。

NOW I ASK YOU · 質問します

Four questions for you

みなさんへの4つの質問

1 Where did YOUR eyes go first, on the very first picture? Why?

最初の写真で、あなたの目は最初にどこへ行きましたか？なぜ？

2 What is the most “salient” thing in this classroom?

この教室で一番「目立つ」ものは何ですか？

3 If you could scan ONE object in Japan in 3D and keep it forever, what would it be?

日本の物を一つ3Dスキャンして永遠に残せるなら、何にしますか？

4 Which country would you like to visit to study science?

科学を学ぶためにどの国へ行きたいですか？

WORDS answer 答え Japanese is fine 日本語でもOK

QUESTIONS AND ANSWERS · 質疑応答

What would you like to ask me?

私に何を聞きたいですか？

About triangles, Morocco, my job, or anything at all.

Thank you　ありがとうございました　شكرا　ⵜⴰⵏⵎⵎⵓⵜ　Merci

Zaineb IBORK, Ph.D. · JSPS Postdoctoral Fellow · Computer Vision and Discrete Geometry Group, NII

REFERENCES · 参考文献

Where can you learn more?

もっと知りたい人は、どこで学べますか？

MY RESEARCH · 私の研究

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- Figures: “The many faces of Lara Croft”, infograds.com; AI, machine learning and deep learning, Towards Data Science; Z. Ibork, PhD defence, 2025
- Eye-tracking: X. Ding, Z. Chen, W. Lin, Z. Chen, Towards 3D Colored Mesh Saliency, IEEE TMM 2024; K. Zhang, D. Zhu, X. Min, G. Zhai, Mesh Mamba, CVPR 2025; Textured Mesh Saliency, arXiv, 2024
- 3D datasets: Nehmé et al., IEEE TVCG 27(3), 2021 (CMDM); Chen et al., ACM TOG 31(4), 2012 (Schelling points); Lavoué et al., Computer Graphics Forum 37(2), 2018; Váša & Rus, Computer Graphics Forum 31(5), 2012
- Japan Meteorological Agency (jma.go.jp); Direction Générale de la Météorologie (marocmeteo.ma)
- Photos: Wikimedia Commons (CC BY-SA), NASA/JPL-Caltech, ESA, and personal photos